

The objective of the book is to provide an understanding of the basic causes and effects of ESC in microelectronics, including the physical effects of working under ESD conditions, leading on to the precautions and protection which can or should be taken to ensure reliable circuit performance. The material includes:

- the physics and operation of ESD protection elements
- design and layout considerations
- failure modes and reliability issues
- ESD modelling
- case studies and surrounding aspects.

The emphasis throughout is on practical considerations.

This is a text which will be of interest to many engineers concerned with microelectronic reliability, but of particular use to those beginning or engaged in device research and/or development. The excellent figures in the text and comprehensive end-of-chapter references make it a highly recommended reference book for this rather specialized subject area.

S.L. Hurst

Adaptive Analog VLSI Neural Systems

M.A. Jabri, R.J. Coggins and B.G. Flowers, Chapman and Hall, UK, 1996, 262 pp.

On the back cover it says that this is 'the first practical book on neural networks learning chips and systems', yet in Chapter 1 it mentions not only Carver mead's book *Analog VLSI and Neural Systems*, but also states 'Many books ... have been written ... on analog implementations'. The claim made here is that this is the first *practical* book, and indeed it does describe systems that really have been built and used,

though how many have been seen outside a research laboratory is not clear.

Although I quite liked much of the book, I was not sure who it was aimed at. I, like many other researchers in neural networks, am not really interested in hardware realizations—I am quite happy simulating in software. The main reason for this is that I almost always have to 'modify' standard networks to suit the problem. If I had a hardware realization I could not customize it to my needs, so I think that the people who would benefit most from this book are circuit designers, as this gives them a new field to work in. This being the case, I would have thought that the emphases in Chapters 2 and 3 are the wrong way round. Chapter 2 gives a very brief introduction to neural networks. It does not give much of an explanation of why neural networks are useful, and does not derive any of the equations. The opposite is true for Chapter 3, where nearly 20 pages are devoted to the derivation of the equations and small signal models of a MOSFET. I found this quite daunting and, if I am right about the perceived audience, unnecessary. Having said this, I found that if I just kept going and skimmed the parts that I did not really understand, I could get a good overview of the developments in CMOS technology and the new circuits that have been designed in the last 5 years or so.

The applications were the most interesting parts of the book, and usually there were pieces of information that I found fascinating. The learning mechanisms using weight perturbation, for example, were very nicely described. The book comes to an abrupt end with the Botzmann machine. The description of the theory behind this network was very clearly written, which made this chapter very readable. However, I felt that a concluding chapter with some thoughts on the future of these techniques would have been a nice way to finish the book off.

Book Reviews

As it is, I would think that there will be a small specialist audience out there who will be interested in the book. Unlike the text on the back cover, I would not say that this is a comprehensive introduction to VLSI neural networks. That description is more suited to Carver Mead's book. This book could be described as a useful update on VLSI neural networks.

P.D. Picton

Fourier Transforms: An Introduction for Engineers

R.M. Gray and J.W. Goodman, *Kluwer Academic Publishers, Norwell, MA, USA, 1995, 360 pp., US\$59.95, UK £41.95, Dfl.105.00*

This teaching and reference text covers the basic definitions, properties and examples of use of Fourier transforms and Fourier analysis, including discrete time, continuous time, finite duration and infinite duration signals. In total the aspects which are collectively covered are:

- signals and systems
- the basic definitions of Fourier transforms
- Fourier inversion
- basic properties of the Fourier transform
- generalized transforms and functions
- convolution
- two-dimensional Fourier transforms
- memoryless non-linearities.

Generally the material is very readable, but more worked numerical examples in the text rather than the algebraic developments would be helpful. End-of-chapter problems are given, but no

solutions, which makes them rather ineffective as a teaching means for a reader working on his or her own. Also, for readers specifically interested in digital logic design and test, no coverage is given of the discrete Rademacher, Rademacher-Walsh, Haar or Chrestenson transforms.

In summary, there are possibly other available texts which may be preferred for engineering applications, but this text may be considered as a very useful reference text to refresh memory on all the fundamentals of this mathematical area.

M.S. Harris

Logic and Architecture Synthesis: State-of-the-Art and Novel Approaches

G. Saucier and A. Mignotte (eds), *Chapman and Hall, London, UK, 1995, 383 pp., £55.00*

Fundamental research in the area of logic design tended to be somewhat neglected in the 1980s, due to the rapidly advancing power of the micro-processor and fabrication capabilities for memory and other VLSI circuits. Nevertheless, interest was maintained in certain academic establishments and industrial organizations and has increased in recent years due to the search for more efficient CAD tools and high-level synthesis.

This is, therefore, a timely publication for those engaged in such R&D activities, but not directly useful for design engineers engaged in day-to-day design work. The book explores recent research work involving logic and architectural synthesis, and presents a selection of chapters from selected authors with expertise in specific areas. The majority of the material was presented at the Workshop on Logic and Architecture Synthesis held in Grenoble, France, in December 1994, but has been compiled into stand-alone chapters in this publication.

The text is arranged on 14 topics, namely: